

Ideation Phase

Empathize & Discover

Metric	Details
Date	08 July 2026
Institution	Vishnu Institute of Technology
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It is a useful tool to help teams better understand their users

Creating an effective solution requires understanding the true problem and the person who is experiencing it[cite: 7]. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges

Our Empathy Map Canvas: Clinical System User

Primary user persona: A diagnostic health specialist or cardiovascular informatics consultant working in a busy clinical care center who reviews large multi-variable patient spreadsheets daily to identify early preventative health risks[cite: 4].

THINKS

- Wonders whether undetected risk factors like highBMI, smoking habits, or diabetes are quietly compounding across a patient's history[cite: 4, 5].
- Weighs whether manual spreadsheet crossreferencing is causing critical bottlenecks in patient validation[cite: 4].
- Questions whether generic software layouts arewasting essential screen space instead of focusing on clean clinical metrics[cite: 4].

SEES

- Considers switching platforms if large dataset
- Sees a massive influx of patient diagnosticspreadsheets containing thousands of rows and overlapping fields[cite: 4].
- Notices cluttered UI templates filled withunnecessary distracting corporate branding elements like placeholder text strings[cite: 4, 5].

- Observes clinical trends where high-risk factor clusters instantly emerge when segmenting data by age groups and lifestyle metrics[cite: 4, 5].

- Sees rigid digital layout frameworks that

THINKS

SEES

loading errors or connection timeouts disrupt daily completely lack simultaneous dual-axis view ports hospital workflows.

SAYS

DOES

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- Compares age group cohorts, gender profiles,
 - "Show me exactly how lifestyle inputs like active and diagnostic incident volumes across smoking and alcohol intake correspond directly with interactive panels before confirming an underlying co-morbidities like stroke history." [cite: 4] assessment.
 - "Static medical data spreadsheets are incredibly
 - Filters clinical data parameters dynamically to tedious and time-consuming to read when crossbypass non-essential UI features and isolate referencing multiple attributes." [cite: 4] high-risk groups
- "I need to spot compounding high-risk factor
- Relies repeatedly on clear, custom visual charts clusters across my patient list rapidly before critical to catch latent pathological patterns rather than events take place." [cite: 4] scanning raw database text lines.
 - "I can only make informed, data-driven preventative
 - Interacts with live visualization workspaces and treatment decisions if the core visual markers are contacts engineering support if session highlighted cleanly." [cite: 5] background timeout errors occur.
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PAINS & GAINS

PAINS

- Frustrated when connection timeout disruptions and application sync failures occur when loading large analytics files.
- Disappointed when rigid system boundaries lack simultaneous dual-axis view ports for dashboards alongside diagnostic summaries.
- Finds it tedious to manually search and parse thousands of patient database rows to uncover overlapping demographic and behavioral risks.
- Concerned about cognitive visual overload induced by generic boilerplate template footprints or placeholder corporate text strings.

GAINS

- Values accurate, custom-built, lightweight web dashboard wrappers that load live, responsive analytics seamlessly.
- Appreciates structured diagnostics panels tailored to core health indicators, parameters matrices, and pre-existing medical records.
- Wants confidence that interactive cloud visualizations genuinely reflect comprehensive data pipelines backed by data preprocessing safety checks.
- Enjoys the convenience of navigating cross-filtered demographic, behavioral, and clinical datasets in a single tap.

User Needs

- Multi-variable clinical data predictions grounded in clean, preprocessed health records rather than unorganized rows.
- A dedicated workspace layout that highlights both behavioral habits (smoking/alcohol) and preexisting conditions (diabetes/stroke) concurrently.
- Upfront visibility of system operational metrics, clean parameters scales, and direct navigation links across the frontend interface.
- A clean layout tailored to reducing visual lookup blockages to dramatically accelerate clinical validation timelines.

Opportunities

- Use historical `Heart_new2.csv` data to isolate exact demographic, lifestyle, and co-morbidity risk factor trends.
- Segment patients by age group cohorts and gender criteria to build targeted visual analytics and health risk profiles.
- Correlate patient lifestyle factors with underlying historical data rows to help doctors catch chronic patterns early.
- Highlight peak risk intervals and common medical profiles to guide hospital onboarding, diagnostics, and care management pipelines.

